# PROJECT PLANNING & MANAGEMENT

Store Sales Data Analysis

DEPI Graduation Project • Data Analysis Track

Dataset: Superstore Sales Dataset.csv

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Prepared for project planning and team review

# 1. Project Proposal

## 1.1 Project Overview

The project analyzes the Superstore Sales Dataset to understand sales performance and identify patterns across time, geographic regions, products, and customer segments. The work includes data preparation, analysis, KPI calculation, insight generation, and an interactive dashboard to communicate findings.

## 1.2 Problem Statement

Sales datasets contain information about orders, customers, products, and regions. Without structured analysis, it can be difficult to compare performance, identify trends, and discover useful patterns. This project turns the available records into clear findings that support data-informed decisions.

## 1.3 Project Objectives

- Analyze sales trends over time.
- Compare sales performance across geographic regions.
- Identify products and product categories with the highest sales.
- Analyze sales by customer segment.
- Calculate KPIs supported by the available data.
- Present findings through an interactive dashboard.

## 1.4 Project Scope

In scope: dataset inspection and quality assessment; data cleaning and preparation; analysis by date, region, product, and customer segment; KPI calculation; insight generation; interactive dashboard development; documentation; and GitHub file management.

Scope limitation: profit, profit margin, units sold, and discount amount will not be reported as actual measures unless reliable source fields or supplementary data become available.

# 2. Project Plan

## 2.1 Official Milestones

| Phase | Official Deadline | Expected Deliverables |
|---|---|---|
| Project Planning & Management | 16 Oct 2026 | Planning documents |
| Literature Review | 16 Oct 2026 | Review and improvement notes |
| Requirements Gathering | 16 Oct 2026 | Requirements documentation |
| System Analysis & Design | 6 Nov 2026 | Analysis and design documents |
| Implementation (Source Code & Execution) | 30 Nov 2026 | Prepared data, analysis, dashboard |
| Final Presentation, Testing & Reports | 4 Dec 2026 | Testing records, reports, final presentation |

## 2.2 Proposed Planning Schedule

| Task | Proposed Dates | Output |
|---|---|---|
| Prepare Project Proposal | 10–11 Oct | Proposal draft |

| Task | Proposed Dates | Output |
|---|---|---|
| Prepare Project Plan and Gantt schedule | 11–12 Oct | Timeline and milestones |
| Define roles and responsibilities | 11–12 Oct | Task assignment table |
| Prepare risk plan and KPIs | 12–13 Oct | Risk and KPI sections |
| Review and submit planning documents | 14–16 Oct | Reviewed planning package |

The detailed dates above are proposed for internal organization; the official deadlines are listed separately in the milestones table.

## 2.3 Required Resources

| Resource | Purpose |
|---|---|
| Superstore Sales Dataset.csv | Primary dataset |
| Excel / Power Query | Inspection, cleaning, and preparation |
| Power BI | Analysis and interactive dashboard |
| GitHub | Version control and file sharing |
| Team members and project documentation | Execution, review, and reporting |

## 2.4 Project Monitoring

- Review task progress regularly.
- Track milestones and deliverables.
- Document contributions in GitHub.
- Report delays or blockers early.
- Review output quality before submission.

# 3. Task Assignment & Roles

Team size: 6 members. The role allocation below is a proposed structure and should be confirmed by the team. Replace Member 1–Member 6 with the actual names.

| Member | Role | Responsibilities |
|---|---|---|
| Member 1 | Project Coordinator | Coordinate work, monitor progress, align tasks, and track deadlines. |
| Member 2 | Documentation & Requirements | Prepare and organize project documents; document objectives, scope, requirements, and reports. |
| Member 3 | Data Preparation | Inspect data quality, investigate missing values and duplicates, correct data types, and prepare data for analysis. |
| Member 4 | Data Analyst | Analyze sales, identify patterns, generate insights, and calculate supported KPIs. |
| Member 5 | Dashboard & Visualization | Select suitable visuals and build the interactive dashboard. |
| Member 6 | GitHub & Quality Review | Organize repository files, maintain README, review outputs, and document test findings. |

## Collaboration and Accountability

- Each member is responsible for assigned tasks and documenting contributions.
- Members collaborate on dependent tasks and review one another’s work.
- Files and contributions are maintained in a shared GitHub repository.
- Each member should be able to explain their individual contribution during the final presentation.

# 4. Risk Assessment & Mitigation Plan

| Risk | Potential Impact | Mitigation Strategy |
|---|---|---|
| Missing values | Incomplete information may affect analysis. | Inspect missing values and choose treatment based on the field and analytical need. |
| Incorrect data types | Date and numeric calculations may fail or become inaccurate. | Validate column types and test date conversions before analysis. |
| Postal-code formatting | Leading zeros may be lost if postal codes are treated as numbers. | Store postal codes as text when needed to preserve formatting. |
| Duplicate records | True duplicates may inflate counts or totals. | Investigate duplicates and confirm whether records are genuinely duplicated before removal. |
| Incorrect calculations | KPIs and visuals may be inaccurate. | Validate formulas and reconcile totals with the source data. |
| Missing profit or quantity fields | Some profitability and unit-based KPIs cannot be calculated directly. | Do not present these as actual measures without reliable source or supplementary data. |
| Task delays | Project milestones may be missed. | Track deadlines, assign owners, and report delays early. |
| Communication issues | Tasks may overlap or remain incomplete. | Clarify responsibilities and hold regular progress check-ins. |
| GitHub conflicts | Changes may be overwritten or conflict. | Use clear commit messages and review changes before merging. |
| Dashboard errors | Visuals may misrepresent results. | Test measures, filters, and totals before submission. |

## Risk Monitoring

Review data quality before analysis, record issues and corrective actions, validate metrics and visuals, and update the risk plan when new issues arise.

# 5. Key Performance Indicators (KPIs)

The KPIs below are proposed for the project. Some are headline metrics, while others are analytical breakdowns for charts and comparisons.

| No. | KPI | Description | Calculation |
|---|---|---|---|
| 1 | Total Sales | Total recorded sales value. | SUM(Sales) |
| 2 | Sales Growth Rate (%) | Sales change compared with a previous period. | (Current Sales − Previous Sales) ÷ Previous Sales × 100 |
| 3 | Total Orders | Number of unique orders. | DISTINCTCOUNT(Order ID) |
| 4 | Average Order Value (AOV) | Average sales value per unique order. | Total Sales ÷ Total Orders |
| 5 | Total Customers | Number of unique customers. | DISTINCTCOUNT(Customer ID) |
| 6 | Repeat Customer Rate (%) | Share of customers with more than one order in a defined period. | Repeat Customers ÷ Total Customers × 100 |
| 7 | Average Orders per Customer | Average unique orders per customer. | Total Orders ÷ Total Customers |
| 8 | Sales per Customer | Average sales per unique customer. | Total Sales ÷ Total Customers |
| 9 | Average Daily Sales | Average sales per day in the selected period. | Total Sales ÷ Number of days in period |
| 10 | Sales by Region | Compare sales across regions. | SUM(Sales) by Region |
| 11 | Sales by Category | Compare sales across product categories. | SUM(Sales) by Category |
| 12 | Sales by Sub-Category | Compare sales across sub-categories. | SUM(Sales) by Sub-Category |
| 13 | Sales by Product | Compare sales by product. | SUM(Sales) by Product |
| 14 | Sales by Segment | Compare sales across customer segments. | SUM(Sales) by Segment |
| 15 | Sales by Ship Mode | Compare sales by shipping mode. | SUM(Sales) by Ship Mode |

## Implementation Notes

• Total Sales, Sales Growth Rate, Total Orders, Average Order Value, and Total Customers are candidates for headline KPI cards.
• Regional, product, sub-category, segment, and shipping-mode measures are generally better shown as charts or comparisons.
• Sales Growth Rate requires a defined comparison period and non-zero previous-period sales.
• Repeat Customer Rate requires a defined time window and a clear repeat-customer definition.
• Average Daily Sales must specify whether the denominator includes all calendar days or only selling days.

## Data Limitations

The current dataset does not provide dedicated Profit, product-cost, Quantity, or Discount fields. Therefore, Total Profit, Profit Margin, Units Sold, and Discount Amount cannot be calculated reliably from this file alone. They should be added only if trustworthy supplementary data becomes available.

## Final Submission Checklist

☐ Project Proposal
☐ Project Plan and schedule
☐ Task Assignment & Roles
☐ Risk Assessment & Mitigation Plan
☐ KPIs
☐ Review formatting and consistency
☐ Upload project documents to GitHub

Official Planning & Management deadline: 16 October 2026.